NICHOL FLOWERS

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SUMMARY

Data Science professional specializing in Machine Learning, Generative AI, Statistics and Data Engineering. Experience developing and evaluating models, designing experiments and applying quantitative analysis to complex problems. Combines technical capabilities with extensive experience developing metrics, measuring performance, identifying drivers of outcomes and translating findings into measurable operational improvements.

EDUCATION

UC Berkeley - Master of Information and Data Science (MIDS)
Texas Southern University - Bachelor of Science, Chemistry

TECHNICAL SKILLS

Python, SQL, R, Generative AI, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Engineering, LLM Evaluation, LangChain, Qdrant, RAGAS, Hugging Face, Machine Learning, Deep Learning, Neural Networks, TensorFlow, Computer Vision, Reinforcement Learning, Model Evaluation, Data Engineering, Neo4j, AWS, Docker, Git, Data Visualization, Statistics, D3.js, Tableau

PROJECTS

RAG for Multiple Audiences, UC Berkeley School of Information

- Built a dual-audience RAG pipeline in LangChain and evaluated embeddings, chunking strategies, retrieval configurations, reranking and generative models.
- Developed an evaluation framework using LLM-as-a-Judge and retrieval-based metrics.
- Balanced evaluation rigor against token consumption and operational cost during large-scale testing.

Eye2Voice: Gaze-to-Speech, UC Berkeley School of Information

- Developed a real-time gaze-to-speech system enabling individuals with severe physical disabilities to communicate using eye movements.
- Applied deep learning and computer vision techniques to translate gaze patterns into spoken output.
- Awarded the **Hal R. Varian Award** for top Data Science Capstone project.

Refugee Movement Network Analysis, UC Berkeley School of Information

- Analyzed a Neo4j graph of global refugee flows from United Nations refugee data with countries as nodes and refugee movements as directed weighted relationships.
- Applied degree centrality to identify the top refugee-sending and receiving countries in 2010, 2015 and 2020 and visualized incoming and outgoing flows in Python.
- Mapped connections from the top origin countries to asylum countries across the three years using minimum spanning trees.

Predicting California Wildfires, UC Berkeley School of Information

- Built an 8-class wildfire-occurrence classifier spanning 3-, 7- and 14-day forecast horizons over 40+ years of California weather and fire data.
- Engineered a geospatial dataset with 100+ features by merging weather, fire-perimeter and climate-division data with GeoPandas, including lagged temporal features.
- Tuned XGBoost with temporal validation and achieving up to 4x the accuracy over a random baseline.

PROFESSIONAL EXPERIENCE

Cloud Engineer - Problem Management *2021 - Present*
Cisco Systems Remote

- Lead development of LLM tooling that synthesizes unstructured incident data into a structured incident analysis and generates targeted retrospective questions, cutting retrospective meeting preparation from 5–7 days to under 4 hours.
- Design prompts and guardrails that shape how the LLM writes and reasons, including enforcing rules for verbosity, tone and clarity, avoiding questions about actions already taken and flagging failed extractions to users.
- Facilitate Webex incident retrospectives, leading discussions with service engineering teams to drive corrective actions that prevent recurrence or reduce service impact.

Volunteer Research Associate *2017 - 2018*
University of Illinois - Chicago Chicago, IL

- Improved data quality and consistency for federally funded nutrition and public health research.

Global Problem Manager *2012 - 2017*
HERE Technologies (formerly Navteq & Nokia) Chicago, IL

- Developed operational performance metrics and targets using historical trends and customer requirements to measure service performance and identify improvement opportunities.
- Led the automation of Problem Management reporting, developing real-time dashboards to track performance against defined metrics and targets.
- Improved IT service availability by 35.8% by reducing the volume and impact of service disruptions through root cause analysis.

Process Improvement Lead *2010 - 2012*
Navteq (formerly Nokia, currently HERE) Chicago, IL

- Pioneered Navteq's first product-specific strategy for map data collection and updates, contributing to $8.5 million in cost savings across the product portfolio.
- Optimized map change detection processes reducing manual data collection time by 50%.